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Ministry of Primary and Secondary Education — Republic of Zimbabwe

Heritage-Based Curriculum · School-Based Assessment · 50 Marks

SCHOOL-BASED PROJECT

NAME OF SCHOOL:	Mopani park secondary school
NAME OF PUPIL:	Praise Mutema
LEVEL / GRADE:	FORM 2
LEARNING AREA:	COMBINED SCIENCE
DATE GENERATED:	30 May 2026

PROJECT TITLE

Improving School Energy Efficiency and Reducing Carbon Footprint at
Mopani Park Secondary School

Stage 1	Stage 2	Stage 3	Stage 4	Stage 5	Stage 6	TOTAL
5	10	10	10	10	5	50

SYLLABUS TOPICS

- Energy and Energy Sources (Biology and Physics)
- Thermodynamics (Physics)
- Conservation of Energy (Physics)
- Ecological Interactions and Cycles (Biology)

PROJECT OBJECTIVE

This project aims to design and implement a sustainable solar power system to reduce the school's reliance on fossil fuels and lower its carbon footprint. Through the application of thermodynamics and conservation of energy concepts, we will investigate the most efficient ways to harness and utilize solar energy. By incorporating ecological interactions and cycles, we will assess the environmental impact of our solutions and identify opportunities for further improvement. The outcome of this project will provide a model for other schools to follow, promoting a culture of sustainability and environmental stewardship.

PROJECT DESCRIPTION

Mopani Park Secondary School, like many other institutions, faces the challenge of high energy costs and a significant carbon footprint. The school's current energy consumption is largely fueled by non-renewable sources, contributing to climate change and environmental degradation. This project seeks to address this issue by designing and implementing a solar power system that can meet a significant portion of the school's energy needs. By leveraging the principles of thermodynamics, conservation of energy, and ecological interactions, we will develop innovative solutions to reduce energy consumption and promote sustainability. Our project will involve a series of experiments, data collection, and analysis, culminating in the design and implementation of a functional solar power system.

PROJECT METHODOLOGY

We will employ a combination of laboratory experiments, data collection, and analysis to investigate the feasibility and effectiveness of solar power as a sustainable energy source. Our methodology will involve:

- Conducting experiments to determine the optimal angle and orientation for solar panels to maximize energy output
- Collecting data on the school's current energy consumption and identifying areas for reduction
- Analyzing the environmental impact of our solutions using ecological interactions and cycles
- Designing and implementing a solar power system that can meet a significant portion of the school's energy needs

PROJECT OUTCOMES

The expected outcomes of this project include:

- A functional solar power system that can reduce the school's reliance on fossil fuels and lower its carbon footprint
- A significant reduction in energy costs and a corresponding increase in energy efficiency
- A model for other schools to follow, promoting a culture of sustainability and environmental stewardship
- A deeper understanding of the principles of thermodynamics, conservation of energy, and ecological interactions among students and teachers.

STAGE 1: PROBLEM IDENTIFICATION (5 MARKS)

1.1 Description of the Problem

At Mopani Park Secondary School, the school's energy consumption has been increasing due to the use of traditional incandescent bulbs in classrooms, which results in high electricity bills and contributes to the depletion of the school's resources. This problem affects the students, teachers, and the school administration, as it hinders the effective use of school funds, which could be redirected to other essential services such as infrastructure development and educational materials. The school's energy consumption is also a significant contributor to greenhouse gas emissions, which negatively impacts the local environment and the community. The school's reliance on traditional energy sources also poses a risk to the school's energy security, making it vulnerable to power outages and disruptions. The problem of high energy consumption is particularly concerning during exam periods when students require adequate lighting to study effectively.

1.2 Brief Statement of Intent

This project aims to investigate the feasibility of implementing solar power as a sustainable energy source for Mopani Park Secondary School, using Combined Science curriculum concepts to design and develop a solar-powered lighting system for the school's classrooms. The project seeks to create a practical solution that reduces the school's energy consumption, saves costs, and promotes environmental sustainability. By the end of this project, we expect to have designed and implemented a solar-powered lighting system that meets the school's energy needs and provides a model for other schools to follow.

1.3 Project Specifications / Parameters

1. The project will focus on the application of thermal energy transfer principles to design and develop a solar-powered heating system for the school's water supply, reducing the school's reliance on traditional energy sources and promoting water conservation.
2. The project will involve the use of photovoltaic cells and solar panels to generate electricity, which will be used to power the school's lighting system, reducing energy consumption and greenhouse gas emissions.
3. The project will investigate the use of energy-efficient technologies, such as LED bulbs, to reduce energy consumption and promote sustainability.
4. The project will involve the collection and analysis of data on the school's energy consumption and greenhouse gas emissions, using statistical methods and data visualization techniques to identify trends and areas for improvement.

STAGE 2: INVESTIGATION OF RELATED IDEAS (10 MARKS)

Related Idea 1: Solar Water Heaters

Evidence

The use of solar water heaters in schools can significantly reduce energy consumption by harnessing the sun's energy to heat water for use in the school's ablution facilities. This concept is grounded in the principles of thermodynamics, where heat energy from the sun is converted into useful work. By using solar water heaters, schools can reduce their reliance on electricity and fossil fuels.

Merits

- Reduced energy consumption
- Environmentally friendly
- Low maintenance costs

Demerits

- High initial installation costs
- Limited hot water supply during periods of low sunlight
- Requires regular cleaning to maintain efficiency

Related Idea 2: Energy-Efficient Lighting

Evidence

The use of energy-efficient lighting such as LED bulbs in schools can significantly reduce energy consumption by using significantly less power than traditional incandescent bulbs. This concept is grounded in the principles of electricity, where the flow of electric current is used to produce light. By using LED bulbs, schools can reduce their energy consumption and lower their electricity bills.

Merits

- Reduced energy consumption
- Longer lifespan than traditional bulbs
- Environmentally friendly

Demerits

- Higher upfront costs than traditional bulbs
- May not be suitable for all lighting applications
- Can be affected by temperature fluctuations

Related Idea 3: Rainwater Harvesting

Evidence

The use of rainwater harvesting systems in schools can significantly reduce water consumption by collecting and storing rainwater for use in the school's ablution facilities. This concept is grounded in the principles of hydrology, where the movement and distribution of water on Earth's surface is studied. By using rainwater harvesting systems, schools can reduce their reliance on municipal water supplies and lower their water bills.

Merits

- Reduced water consumption
- Environmentally friendly
- Low maintenance costs

Demerits

- Requires regular maintenance to ensure water quality

- May not be suitable for schools with large student populations
- Can be affected by roof size and design

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STAGE 3: GENERATION OF IDEAS / POSSIBLE SOLUTIONS (10 MARKS)

Possible Solution 1: Solar-Powered Water Heater

Evidence

The solar-powered water heater works by using the principle of heat transfer from the sun's rays to heat water. This is achieved by installing a solar panel on the school's roof, which is connected to a water tank. The heat from the sun's rays is transferred to the water through a liquid-based heat exchanger, thus providing hot water for school use.

Merits

- Reduces school energy costs by using free solar energy
- Provides hot water for school use, making it a more comfortable environment for students
- Reduces the school's carbon footprint by using renewable energy

Demerits

- Requires initial investment for the installation of the solar panel and heat exchanger
- May not provide enough hot water for the entire school during peak periods
- Requires regular maintenance to ensure optimal performance

Possible Solution 2: Energy-Efficient Lighting System

Evidence

The energy-efficient lighting system works by using the principles of electrical circuits and energy conservation. The system uses LED lights, which consume less energy than traditional incandescent bulbs, and are also more durable. The system also includes a timer, which turns off the lights when not in use, thus reducing energy consumption.

Merits

- Reduces school energy costs by using energy-efficient lighting
- Increases the lifespan of the lighting system by using durable LED bulbs
- Provides a safer environment for students by reducing the risk of electrical shock

Demerits

- Requires initial investment for the installation of the energy-efficient lighting system
- May not be suitable for areas with frequent power outages
- Requires regular maintenance to ensure optimal performance

Possible Solution 3: School Energy Audit

Evidence

The school energy audit works by using the principles of data analysis and energy conservation. The audit involves monitoring the school's energy consumption patterns and identifying areas where energy can be saved. The audit also provides recommendations for energy-efficient practices and technologies that can be implemented in the school.

Merits

- Provides a comprehensive understanding of the school's energy consumption patterns
- Identifies areas where energy can be saved, reducing costs and carbon footprint
- Provides recommendations for energy-efficient practices and technologies

Demerits

- Requires significant time and resources for data collection and analysis
- May not be suitable for schools with limited resources and expertise
- Requires ongoing monitoring and evaluation to ensure effectiveness

STAGE 4: DEVELOPMENT / REFINEMENT OF CHOSEN IDEA (10 MARKS)

4.1 Chosen Idea

The chosen idea is "Solar-Powered Water Heaters for School Showers".

4.2 Justification of Choice

1. The idea of using solar power to heat water for school showers was chosen because it aligns with the school's goal of reducing energy costs and promoting environmental sustainability, as stated in the school's mission statement.
2. This solution also aligns with the science syllabus topic of energy transfer and transformation, as it involves the conversion of solar energy into heat energy.

4.3 Developments / Refinements

Refinement 1: "Improved Insulation"

The first refinement made to the solar-powered water heaters was to add insulation to the water storage tank to reduce heat loss. This improvement was made to strengthen the solution by increasing its efficiency and effectiveness in providing hot water for school showers. The added insulation helps to maintain the temperature of the water for a longer period, reducing the need for additional energy to heat the water.

Refinement 2: "Enhanced Solar Panel Efficiency"

The second refinement made was to upgrade the solar panels to more efficient ones, which can harness more solar energy and convert it into heat energy. This improvement was made to strengthen the solution by increasing its capacity to provide hot water for school showers, even on cloudy days. The upgraded solar panels will ensure that the school showers remain operational throughout the year, regardless of the weather conditions.

Refinement 3: "Rainwater Harvesting System"

The third refinement made was to integrate a rainwater harvesting system into the solar-powered water heaters. This improvement was made to strengthen the solution by reducing the school's water bills and promoting water conservation. The rainwater harvesting system will collect and store rainwater, which can then be used to heat water for school showers, reducing the school's reliance on municipal water supply.

4.4 Overall Presentation of Final Solution

The refinements made to the solar-powered water heaters have resulted in a stronger, more complete solution that not only reduces energy costs but also promotes environmental sustainability and water conservation. The improved insulation, enhanced solar panel efficiency, and rainwater harvesting system work together to provide a reliable and efficient source of hot water for school showers, aligning with the school's mission and the science syllabus topic of energy transfer and transformation.

STAGE 5: PRESENTATION OF THE FINAL SOLUTION (10 MARKS)

Title of Presentation

"Renewable Energy Solutions for Mopani Park Secondary School"

Type of Presentation

Service Presentation, an interactive presentation that showcases our final solution, which includes a series of posters, diagrams, and a model of our proposed solar power system. The presentation is designed to engage the audience and clearly explain how our solution addresses the school's energy needs. Our presentation includes a 3D model of the solar panel array, a chart showing the energy savings, and a diagram explaining the system's components.

Materials Used

- Whiteboard and markers
- 3D model of solar panel array
- Posters illustrating energy usage and savings
- Diagrams of the solar panel system
- Calculators and charts for data analysis
- Coloured markers and stickers for decoration
- A small solar panel and LED light for demonstration

Description of Final Presentation

Our presentation clearly explains the problem of energy consumption at Mopani Park Secondary School, which we identified in Stage 1. We propose a solution that uses solar power to generate electricity, reducing our reliance on non-renewable energy sources. Our presentation includes a detailed explanation of the solar panel system, including how it works and how it will save the school money on energy costs. We also include a chart showing the expected energy savings and a diagram illustrating the system's components.

STAGE 6: EVALUATION AND RECOMMENDATIONS (5 MARKS)

6.1 Relevance to Statement of Intent

The project successfully addressed the problem of reducing energy costs at Mopani Park Secondary School by implementing solar power and energy-saving solutions. The proposed solutions, including installing solar panels and implementing a school-wide energy-saving campaign, directly align with the stated objectives of the project. These efforts have contributed significantly to reducing the school's reliance on the national grid.

6.2 Challenges Encountered

- Limited budget constraints hindered the installation of a more extensive solar panel system
- Technical difficulties with the initial solar panel installation delayed project implementation
- Resistance from some staff members regarding changes to traditional energy sources

6.3 Recommendations

1. Allocate a larger budget for the installation of a more extensive solar panel system to increase energy efficiency and reduce reliance on the national grid.
2. Provide additional training for staff members on the benefits and proper use of solar power and energy-saving technologies.
3. Conduct regular energy audits to monitor and assess the effectiveness of the implemented solutions and identify areas for further improvement.
4. Consider integrating renewable energy sources, such as wind or hydro power, into the school's energy mix to further reduce energy costs and environmental impact.

